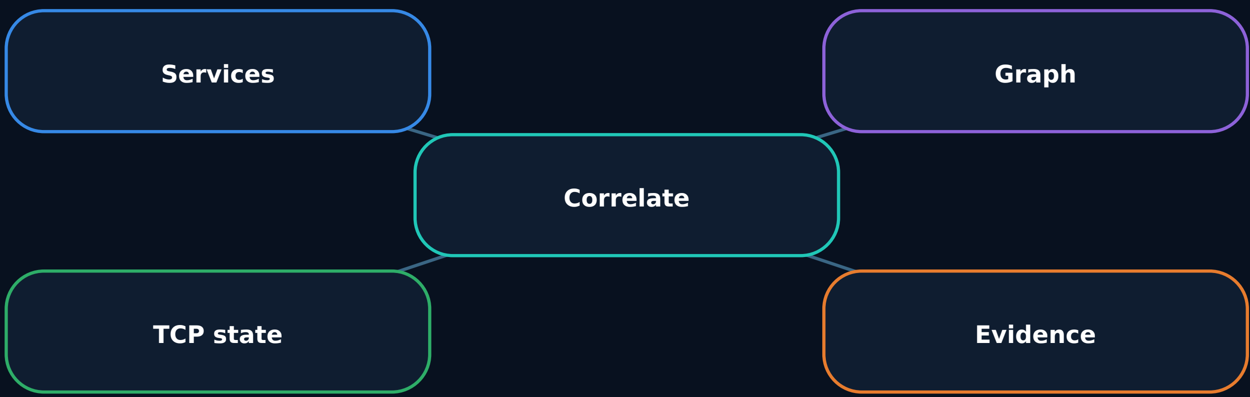
SERVICE DEPENDENCY MAPPER

User Manual

A complete guide to installing, running, interpreting and maintaining the lightweight local Windows service dependency scanner.

Release 2.0.1 | Windows PowerShell 5.1 | 29 August 2026



LOCAL ONLY	READ ONLY	ONE HTML REPORT

Contents

Use this manual as both a first-run guide and the ongoing operating reference for Service Dependency Mapper v2.0.1.

> **Manual conventions**
>
> Commands appear in dark code blocks. Blue callouts explain normal behaviour, green callouts confirm safety boundaries, orange callouts identify interpretation risks, and red callouts identify actions that should stop until reviewed.

Audience

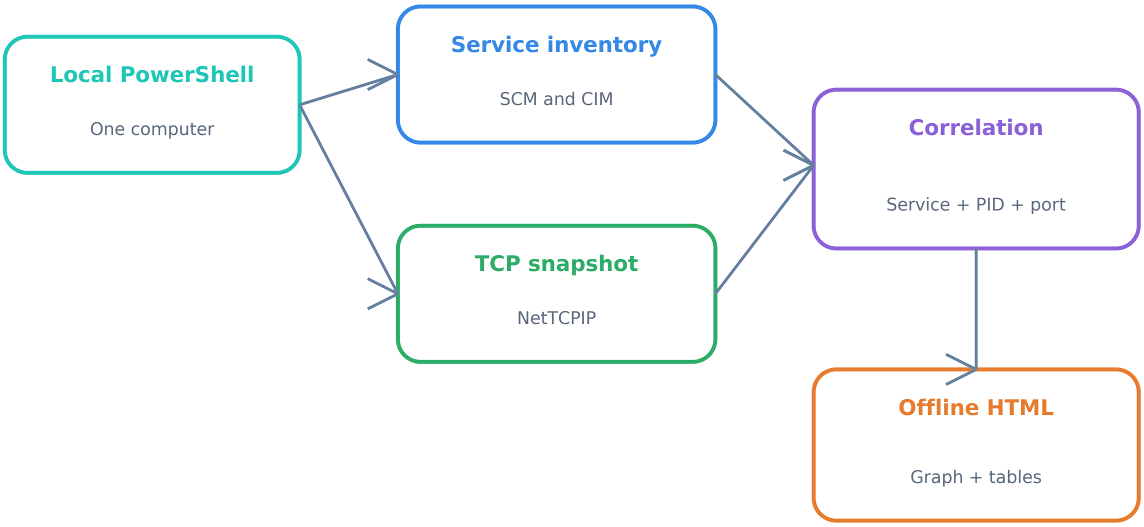
This manual is written for Level 1 engineers, server administrators, application support teams and maintainers. No prior knowledge of the script internals is required for normal operation.

1. Product overview

Service Dependency Mapper v2.0.1 is a lightweight PowerShell application that runs directly on the Windows computer being inspected. It combines service-control data with a point-in-time TCP snapshot and writes one self-contained interactive HTML report.

> **Core outcome**
>
> The report answers: Which services exist? What do they declare as dependencies? Which ports do their processes listen on? Which current TCP endpoints do those processes connect to?

Component model



The complete run stays on one computer. Collection, correlation, HTML generation and report viewing do not require a remote server or internet service.

Requirements

Requirement	Minimum / expected state	Why it is needed
Operating system	Windows 10/11 or Windows Server	Provides the service, CIM and TCP data sources.
PowerShell	Windows PowerShell 5.1	Runs the self-contained script and WinForms menu.
Privileges	Administrator recommended	Improves access to service paths and TCP process ownership.
Browser	A current local browser	Opens the generated offline HTML report.
Additional software	None	No Python, module, agent, web server or paid component is used.

Output

```
Reports\Service-Dependency-Report-SERVER01-YYYYMMDD-HHMMSS.html
```

2. Safety, permissions and data handling

> **Read-only design**
>
> The collector reads operating-system state and writes only to the selected report folder. It does not stop, start, restart, install, remove or reconfigure a Windows service.

What the tool reads

Source	Information read
Win32_Service	Name, display name, state, start mode, service account, PID, binary path and description.
Get-Service	Declared dependencies and services that depend on each service.
Get-NetTCPConnection	Current TCP listeners and connections, addresses, ports, state and owning PID.
Get-NetIPAddress	Local IPv4 and IPv6 addresses used to recognise local TCP relationships.
Current identity	Whether the PowerShell process is running as an administrator.

What the tool does not change

Area	Explicit boundary
Services and processes	No service control, process termination or executable change.
Network and firewall	No firewall rule, adapter, route, DNS or TCP configuration change.
Management	No WinRM, PowerShell remoting, scheduled task or registry modification.
Execution policy	The launcher requests Bypass only for its child process. It does not call Set-ExecutionPolicy.
External communication	The generated report contains no CDN, analytics, external font or API request.

Administrator elevation

A standard-user run is permitted. The tool records a warning and continues, but Windows may withhold some binary paths or ownership information. For production evidence, run the launcher with Run as administrator.

Execution policy

The CMD launcher uses `-ExecutionPolicy Bypass` for one PowerShell process. Domain-enforced MachinePolicy or UserPolicy still takes precedence. Do not weaken Group Policy to run this tool; use the approved signing or allow-listing process.

> **Sensitive output**
>
> The HTML report can expose server names, service accounts, executable paths, ports and remote addresses. Store, transmit and retain it as sensitive infrastructure information.

3. Installation and first run

There is no installer. Keep the five release files together in one controlled folder.

Step 1 - Extract the release

```
C:\Tools\Service-Dependency-Mapper
```

Step 2 - Confirm the files

File	Purpose
ServiceDependencyMapper.ps1	Collector, correlation logic, WinForms menu and HTML generator.
Start-ServiceDependencyMapper.cmd	Double-click launcher with process-scoped Bypass and STA mode.
README.md	Technical operating and maintenance reference.
QUICKSTART.txt	Short first-line operating instructions.
LICENSE	MIT licence.

Step 3 - Start with elevation

Right-click Start-ServiceDependencyMapper.cmd and select Run as administrator. If Windows displays a downloaded-file or SmartScreen prompt, follow your organisation's approved validation and allow-listing procedure.

Step 4 - Use the recommended defaults

- Keep the default `Reports` folder or choose a protected local folder.
- Leave Include stopped services selected for the complete service-control map.
- Leave Open the HTML report selected for the first run.
- Click Scan this computer and keep the window open until completion.

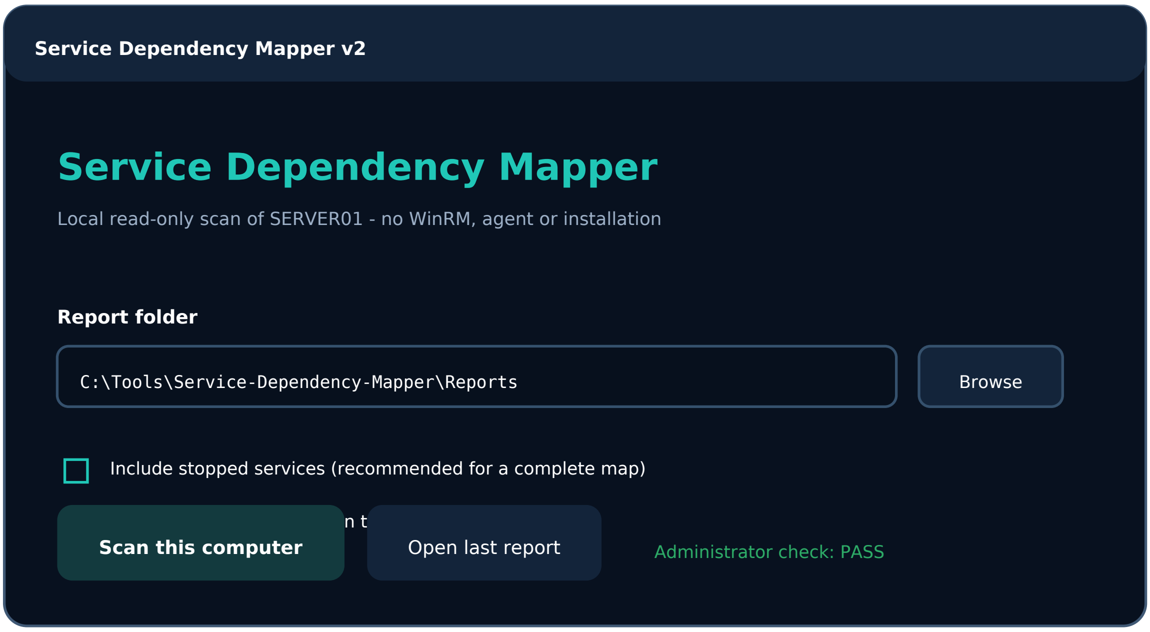
Step 5 - Confirm successful output

- A completion dialog displays the report path.
- The report opens in the default browser when that option is enabled.
- The filename contains the local computer name and collection timestamp.

> **Expected first-run time**
>
> Most endpoints complete quickly. Runtime depends on service count, TCP state and local CIM responsiveness; the tool performs no long observation window or packet capture.

4. WinForms menu walkthrough

The default launcher opens a deliberately small menu. There is no server-name field because v2 is local-only.



Illustrative layout of the bundled WinForms menu.

Control	How to use it
Report folder	Choose where the HTML file will be written. The folder is created if it does not exist.
Browse	Select an existing folder using the standard Windows folder dialog.
Include stopped services	Enabled by default. Clear it only when a running-services-only view is specifically required.
Open the HTML report	Starts the completed report with the default browser.
Administrator check	Green PASS indicates elevation. A warning allows collection but signals possible partial data.
Scan this computer	Runs service collection, TCP collection, correlation and report generation.
Open last report	Reopens the latest report created during the current menu session.

During collection

The form shows a moving progress indicator and disables scan and browse controls to prevent a second run from starting. The scan itself runs in the same local PowerShell process. If an exception occurs, the menu displays the error and re-enables the controls.

> **Server Core behaviour**
>
> If WinForms cannot load, the script falls back to console mode and prints the generated report path. For predictable Server Core use, invoke the script with -NoGui.

5. Console operation and parameters

Console mode is useful for Server Core, repeatable local checks, troubleshooting the menu, or launching from another approved local automation wrapper.

Open the GUI explicitly

```
powershell.exe -NoLogo -NoProfile -ExecutionPolicy Bypass -STA `
    -File .\ServiceDependencyMapper.ps1
```

Run a complete local scan

```
powershell.exe -NoLogo -NoProfile -ExecutionPolicy Bypass `
    -File .\ServiceDependencyMapper.ps1 `
    -NoGui `
    -OpenReport
```

Run only running services into another folder

```
powershell.exe -NoLogo -NoProfile -ExecutionPolicy Bypass `
    -File .\ServiceDependencyMapper.ps1 `
    -NoGui `
    -RunningServicesOnly `
    -OutputFolder C:\Temp\ServiceReports `
    -OpenReport
```

Parameter reference

Parameter	Default	Effect
-OutputFolder	Reports beside script	Selects the local output folder.
-NoGui	Off	Skips WinForms and runs collection immediately.
-RunningServicesOnly	Off	Excludes stopped services from the service inventory and source-service map.
-OpenReport	Off in console	Opens the completed HTML report.

Console result

A successful console run prints the computer name, service count, relationship count, warning count, report path and duration. It also returns a PowerShell object with the same summary for a calling wrapper.

> **Local-only enforcement**
>
> There is no ComputerName parameter. The target is always $env:COMPUTERNAME in the current Windows session.

6. How the collector works

The scan runs as a fixed local pipeline. Each stage supplies a small, specific data set to the next stage.

Stage	Function	Work performed
1	Test-SdmAdministrator	Checks whether the current token belongs to the local Administrators role.
2	Get-SdmServiceInventory	Reads CIM service fields and Service Controller dependency properties.
3	Get-SdmNetworkSnapshot	Reads current TCP state and local IP addresses when NetTCPIP cmdlets exist.
4	New-SdmRelationships	Builds service, PID, listener and endpoint relationships with confidence labels.
5	New-SdmHtmlReport	HTML-encodes values, creates graph/evidence markup and writes one UTF-8 HTML file.

Service inventory fields

Field	Source / meaning
Name and DisplayName	Internal service key and friendly Windows display label.
State and StartMode	Current running/stopped state and configured automatic/manual/disabled mode.
StartName	Account configured to run the service.
ProcessId	Current hosting PID; stopped services normally report 0.
PathName	Service binary/command path with common password, token, secret and API-key patterns redacted.
Dependencies	ServicesDependedOn from the Service Controller object.
Dependents	DependentServices from the Service Controller object.

Index construction

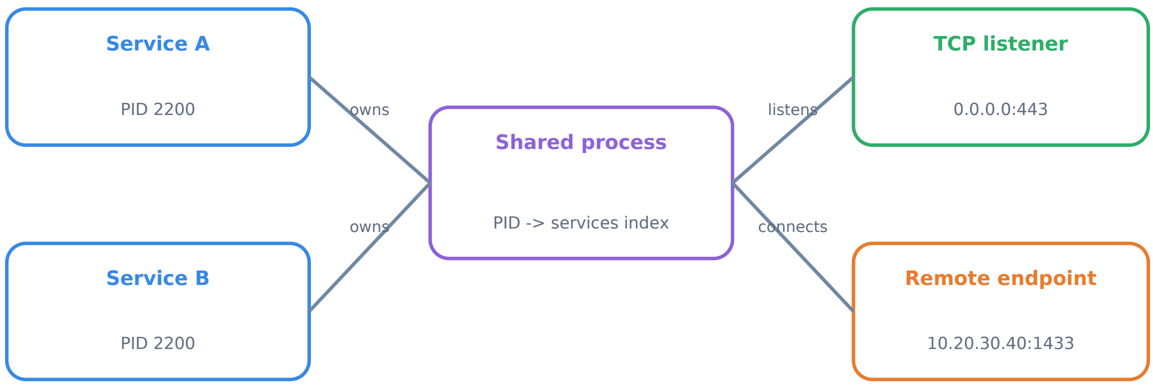
- A service-name index supports fast lookup of friendly target labels.
- A PID-to-services index connects current TCP ownership to running services.
- A listener-by-port index supports local service-to-service TCP correlation.
- A relationship key prevents duplicate source/type/target/direction rows.

Failure handling

Service collection is essential; if no services are returned, the run stops. TCP collection is optional; if the cmdlet is missing or access fails, the report is still created with service-control dependencies and a warning.

7. TCP and PID correlation

Windows TCP records identify an owning process, not a service. The mapper uses each service's current PID to bridge that gap.



Both services receive Shared PID confidence because Windows hosts them in the same process.

Shared-process attribution is intentionally visible rather than silently presented as exact.

Listener correlation

- Connections with state `Listen` are formatted as local address and port.
- The owning PID is looked up in the PID-to-services index.
- Every service hosted by that PID receives a Listening port relationship.

Outbound correlation

- Non-listening records with a positive remote port are treated as current outbound evidence.
- The owning PID identifies the candidate source service or shared set of services.
- A remote address and port become an Outbound TCP endpoint unless local correlation succeeds.

Observed local TCP correlation

- The remote address is compared with the computer's local IPv4, IPv6 and loopback addresses.
- The remote port is compared with local listener ports.
- When a matching listener PID maps to a service, the endpoint becomes an Observed local TCP service relationship.

Confidence labels

Label	Meaning
Direct PID	Exactly one scanned service uses the owning PID.
Shared PID (N services)	Several services use the PID; the network edge cannot be attributed to only one of them.
Declared	The relationship came from Service Control Manager dependency data, not network inference.

8. Relationship model

Every report edge has a source service, target, relationship type, direction, evidence string and confidence label.

Relationship	Direction	Target	Evidence basis
Declared dependency	Selected service -> target	Service	The selected service lists the target in ServicesDependedOn.
Dependent service	Target -> selected service	Service	Another service lists the selected service as its dependency.
Listening port	Selected service -> listener	IP address and TCP port	A current Listen record is owned by the service PID.
Outbound TCP	Selected service -> endpoint	Remote IP address and port	A current non-listening TCP record is owned by the service PID.
Observed local TCP	Selected service -> local service	Service	A local remote-address/port pair matches a listener PID belonging to a service.

Direction in the graph

Arrow direction reflects dependency direction. A Declared dependency arrow points from the service that requires something to its target. A Dependent service arrow points from the dependent service towards the selected service. Listener and TCP edges point away from the owning source service.

Evidence strings

Evidence rows include the relevant service name, PID, TCP state, endpoint and listener PID where applicable. Evidence is designed to support review, not to assert business criticality.

> **No automatic verdict**
>
> The tool does not label a service or endpoint safe, unsafe, documented or removable. Operational context must come from application owners and change records.

Deduplication

The script keeps one row for each unique source service, relationship type, target and direction. Repeated TCP records that produce the same logical edge do not flood the report.

9. HTML report overview

The output is one offline HTML file with embedded CSS and JavaScript. It can be copied to another authorised computer and opened directly; no localhost server is required.

Illustrative structure of the generated report.

Header metadata

- Computer name, tool version, scan timestamp and duration.
- Administrator or standard-user status.
- All installed services or running-services-only scope.

Summary cards

Card	Count represented
Services scanned	Number of service inventory records included in the report.
Running	Scanned services whose current state is Running.
Declared dependencies	Unique outbound Declared dependency relationships.
Service listeners	Unique service-to-listener relationships.
Observed TCP dependencies	Outbound TCP plus Observed local TCP relationships.

Built-in report safeguards

- Target-derived text is HTML-encoded before insertion.
- Common secret patterns in service binary paths are redacted before reporting.
- A restrictive Content Security Policy blocks external resources.
- The report contains no external images, fonts, analytics, libraries or network calls.

10. Using the interactive dependency map

The focus map shows the selected service in the centre and its nearby declared or observed relationships around it.

Select a service

- Open the Focus service list and choose a service by display name and internal name.
- The option also shows the total relationship count for that service.
- The service with the most relationships is selected automatically on first load.

Read the nodes

Node colour	Target type	Examples
Blue	Service	RpcSs, EventLog, SQLSERVERAGENT
Green	Listener	0.0.0.0:443, [::]:5985
Orange	Remote endpoint	10.20.30.40:1433
Teal centre	Selected service	The service currently in focus

Read the edges

Each edge label identifies the relationship. Hover over a target node to display the captured evidence in the browser tooltip. Follow the arrow to understand dependency direction.

Twenty-two-node display cap

The graph displays up to 22 relationships for one service to preserve legibility. If more exist, the graph states that it is showing 22 of the total. The evidence table always retains the complete relationship set.

> **Graph versus evidence**
>
> Use the graph to understand shape and the evidence table to make exact technical decisions. A crowded or capped graph is not missing evidence from the underlying report table.

Printing

Use Print / Save PDF in the report header for a static copy. Browser print output is useful for tickets, but the original HTML should be retained when interactive filtering and graph selection are required.

11. Service and dependency tables

The two tables provide the complete searchable evidence behind the visual map.

Service inventory

Column	Interpretation
Service	Friendly display name and internal service name.
State	Current running or stopped state at collection time.
Start mode	Configured automatic, manual or disabled behaviour.
Account	Configured service logon identity.
PID	Current hosting process ID; normally 0 when stopped.
Declared dependencies	Service names required through Service Control Manager.
Binary path	Redacted path and arguments registered for the service.

Dependency evidence

Column	Interpretation
Service	Source display name and internal service name.
Direction	Depends on, Used by, Listens on, Connects locally to or Connects to.
Target	Target service, listener, or remote endpoint.
Relationship	The mapper's exact relationship category.
Confidence	Declared, Direct PID or Shared PID attribution.
Evidence	Plain-language record of the SCM or TCP observation.

Search behaviour

Typing in a table search box hides rows that do not contain the term. Search includes visible fields and the report's stored row-search text. The counter shows visible rows versus the complete table total.

Useful searches

Goal	Example search
Find one service	wuauserv or Windows Update
Find a service account	LocalSystem or DOMAIN\SvcSql
Find an endpoint	10.20.30.40 or 1433
Find shared-process evidence	Shared PID
Find all listeners	Listening port

12. Interpreting results safely

Point-in-time evidence

TCP state is sampled once. A dependency that is idle, scheduled, short-lived or used only during a particular transaction may not appear.

Declared and observed are different

Evidence	What it proves	What it does not prove
Declared dependency	Windows service configuration contains a service-control dependency.	That the dependency is healthy, actively used or sufficient for the application.
Listening port	A service-owned PID was listening at scan time.	That the listener is reachable through firewalls or required by users.
Outbound TCP	A service-owned PID had a current connection.	That the endpoint is approved, permanent or uniquely attributable when the PID is shared.
No observed connection	Nothing matching was visible in this one snapshot.	That the service never uses the endpoint.

Shared PID review

When multiple services use the same process, the tool deliberately assigns the TCP evidence to each candidate service and marks it Shared PID. Review service groups, application documentation, event logs and controlled testing before attributing the connection.

Review checklist

- Confirm the service and application owner.
- Repeat the scan while exercising the relevant application function.
- Compare the timestamp with application and firewall logs.
- Determine whether a hostname, proxy, load balancer or cluster sits behind the observed IP.
- Verify shared-PID evidence with another authorised data source.
- Do not stop a service or block an endpoint solely because it appears unfamiliar.

Change-control rule

Treat the report as discovery evidence attached to a human decision. It is not an automated remediation or firewall-authorisation tool.

13. Recommended operating workflows

A local scan is most useful when paired with a clear question and representative application activity.

Workflow A - First endpoint validation

- Run v2.0.1 locally as Administrator with stopped services included.
- Confirm the summary cards contain service and relationship counts.
- Select several well-known services and verify that their graph and table rows are understandable.
- Record any warnings as expected test limitations, not production findings.

Workflow B - Application dependency review

- Agree the target service names and normal application actions with the owner.
- Start the scan while the application is processing representative work.
- Search the evidence table for those services, ports and destination addresses.
- Validate declared, direct-PID and shared-PID relationships separately.
- Document confirmed dependencies outside the tool in the application record or CMDB.

Workflow C - Before and after an infrastructure change

- Run and retain a timestamped report before the approved change.
- Repeat the same application activity after the change and run another report.
- Compare service rows and endpoint evidence manually or with an approved separate process.
- Record transient differences and seek owner confirmation before treating them as defects.

Workflow D - Incident triage

- Capture the report while the issue is active whenever safe and authorised.
- Use listeners and current outbound endpoints to guide log, firewall and application checks.
- Preserve the original HTML with the incident timeline.

> **Version boundary**
>
> v2 intentionally does not include JSON snapshots or an automated comparison engine. Each run produces a standalone HTML evidence file.

14. Troubleshooting

Symptom	Likely cause	Action
Cannot overwrite variable PID	The original v2.0.0 script used $pid, which collided with read-only $PID.	Replace it with v2.0.1. Confirm the report header shows Tool 2.0.1.
Running scripts is disabled	Execution policy or organisation policy blocks the script.	Use the bundled CMD launcher. If Group Policy still blocks it, follow approved signing or allow-listing; do not weaken policy.
WinForms menu does not open	Server Core, unavailable assemblies or desktop restrictions.	Run with -NoGui and note the report path printed in the console.
No services were returned	CIM failure or insufficient local access.	Run as Administrator; test Get-CimInstance Win32_Service locally.
TCP sections are empty	NetTCPIP cmdlet missing, access denied or no service-owned TCP state visible.	Run Get-Command Get-NetTCPConnection and test an elevated scan.
Many Shared PID rows	Several services are hosted by the same process, commonly svchost.exe.	Treat rows as candidate attribution and verify with other evidence.
Graph shows fewer rows than table	The focus graph caps display at 22 relationships.	Use the complete Dependency evidence table.
Graph is empty for one service	No declared or current observed relationship was captured.	Try a representative activity window and inspect the service table.
Report does not open automatically	Browser association or launch policy prevents Start-Process.	Open the printed .html path manually in an approved browser.
Report folder cannot be written	Permissions, controlled-folder access or invalid path.	Choose a writable protected local folder and rerun.

Useful local checks

```
Get-Command Get-CimInstance, Get-Service, Get-NetTCPConnection
Get-CimInstance -ClassName Win32_Service | Select-Object -First 5
Get-NetTCPConnection | Select-Object -First 5
[Security.Principal.WindowsPrincipal] `
  [Security.Principal.WindowsIdentity]::GetCurrent()
```

Escalation evidence

When escalating, capture the exact message, script version, whether the process was elevated, operating-system edition and whether -NoGui behaves differently.

15. Maintenance guide

All executable behaviour and report markup live in one file: `ServiceDependencyMapper.ps1`. Keep that single-file design unless a real operational requirement justifies added components.

Function ownership

Function	Maintain when
ConvertTo-SdmHtmlText	HTML escaping or supported value rendering changes.
Protect-SdmText	Redaction patterns are reviewed or expanded.
Get-SdmServiceInventory	Service fields or dependency sources change.
Get-SdmNetworkSnapshot	TCP/local-address collection changes.
New-SdmRelationships	PID correlation, relationship types or confidence logic changes.
New-SdmHtmlReport	Report layout, graph, tables or embedded JavaScript changes.
Invoke-SdmLocalScan	Pipeline, output naming or summary behaviour changes.
Show-SdmWinFormsMenu	Menu controls or first-line workflow changes.

Release checklist

- Update $script:SdmVersion and all public documentation together.
- Search assignment targets for collisions with PowerShell automatic variables such as $PID and $Host.
- Review every added cmdlet and prove that it remains read-only.
- Test GUI and -NoGui modes on a non-production Windows endpoint.
- Test elevated and standard-user runs.
- Test a service with no relationship, a direct-PID service and shared-PID services.
- Open the report offline and verify graph selection, searches, tooltips and printing.
- Confirm no external URLs, scripts, fonts, analytics or CDNs were introduced.
- Verify common password/token patterns do not enter the report through PathName.

Minimum test matrix

Test	Expected result
GUI, elevated	PASS status, scan completes, report opens.
Console, elevated	Summary object and report path are printed.
Standard user	Warning recorded; report is created where Windows permits.
No Get-NetTCPConnection	Service-control report is created with a TCP warning.
No active service TCP	Graph/table remain usable with declared dependencies only.
Shared PID	Confidence states Shared PID (N services).

16. Appendices

A. Launcher command

```
powershell.exe -NoLogo -NoProfile -ExecutionPolicy Bypass -STA `
    -File .\ServiceDependencyMapper.ps1
```

B. Output object

Property	Meaning
ComputerName	Local $env:COMPUTERNAME value.
ReportPath	Full path to the generated HTML file.
Services	Number of service inventory records.
RunningServices	Number of scanned records in Running state.
Relationships	Number of unique report relationships.
WarningCount	Number of runtime warnings added to the report.
DurationSeconds	Elapsed scan and report-build duration.
RanAsAdministrator	Boolean elevation result.

C. Glossary

Term	Definition
CIM	Windows management interface used to read Win32_Service.
SCM	Service Control Manager; source of declared service dependencies.
PID	Process identifier that connects current services to TCP ownership.
Listener	A TCP endpoint waiting for inbound connections.
Outbound endpoint	Remote address and port in a current non-listening TCP record.
Shared PID	One process hosting more than one Windows service.
Point-in-time	State visible during one collection rather than historical activity.

D. Operator completion checklist

- I used Service Dependency Mapper v2.0.1 or later on the intended local computer.
- I recorded whether the run was elevated and whether stopped services were included.
- I retained the original HTML securely.
- I reviewed Shared PID and point-in-time limitations, then validated findings with the application or service owner before any change.

End of manual

Service Dependency Mapper v2.0.1 remains intentionally small: one PowerShell script, one launcher and one offline report. Preserve that simplicity as part of its operational value.